

CS112: Wednesday 30 September, 12:15 to 1:20

Monday we wrote `PyList` together. A class holding `Item *myArray`, `mySize` and `myCapacity`; a constructor that sets the pointer to `nullptr`; an `append` that makes room for one more item every time it is called. It compiles, it passes its test, and every line in it is correct.

Today: the object is not the data. `myArray` holds an address, the items sit somewhere else, and that address moves every time the array grows. Two things fall out of seeing that. Our `append` is doing far more work than it needs to, and nothing in the class ever gives the memory back.

TIME	BEAT
12:15	Daily quiz
12:20	1. What does our <code>append</code> cost?
12:32	2. Make it cost less, and ? :
12:48	3. It leaks right now
12:58	4. The debugger, on our own class
13:14	5. What this unlocks

If the clock slips, shorten beat 1. Beat 4 is the one nothing else replaces.

Beat 1. What does our `append` cost? (12 min)

Open `PyList.cpp`. Run `make test`. It passes, as expected. Nothing here is broken. Now let's think about what these operations cost.

Talk to your neighbor. Every `append` asks for a new array and copies everything across. How many item copies is that, for 100 appends?

Collect answers before confirming anything. The first `append` copies nothing, the second copies 1, the third copies 2, and the hundredth copies 99. Put the sum on the board and close it:

$$\text{copies} = \sum_{i=0}^{n-1} i = \frac{n(n-1)}{2} = \frac{100 \times 99}{2} = 4950$$

The closed form is the point, not the number: **n^2 over 2**, so ten times the appends is a hundred times the copying. Measured on our own file: **4,950 item copies and 100 allocations**.

Then hand the problem back rather than the answer:

Talk to your neighbor. The copying is what costs. How could we copy less often?

Whatever comes back, cost it on the board. Two answers are likely:

STRATEGY	100 APPENDS	1000 APPENDS
one more slot each time	4,950 copies	499,500 copies
ten more slots each time	450 copies	49,500 copies
twice the room each time	127 copies	1,023 copies

"Ten extra" is a real improvement and still the wrong shape: ten times the appends, a hundred times the copying. Only doubling's second column barely moves. If nobody reaches it, ask what happens when the extra room is proportional to what we already have instead of a fixed number. Week 5 names why.

Beat 2. Make it cost less, and ? : (16 min)

Write it together, starting from the condition. Ask when the array needs to grow, and take `mySize == myCapacity` from the room.

```
void PyList::append(Item it) {
    if (mySize == myCapacity) {          // full; empty is full too
        int cap = (myCapacity == 0) ? 1 : myCapacity * 2;
        Item *bigger = new Item[cap];
        for (int i = 0; i < mySize; ++i)
            bigger[i] = myArray[i];
        delete [] myArray;
        myArray = bigger;
        myCapacity = cap;
    }
    myArray[mySize] = it;
    ++mySize;
}
```

Monday's `if (mySize == 0) ... else ...` is gone, and that is worth stopping on: **the empty list was never a special case**. It was an array with no room left, which the one condition already catches.

The `? :` line is the only unfamiliar thing on the screen. Put the four-line `if/else` they can already read beside it, and read the one-liner aloud as "if this, then that, otherwise the other".

`if` is a **statement**: it *does* something. `? :` is an **expression**: it *is* a value. That is why it can sit on the right of an `=` and the `if` cannot.

Three rules: choose **between two values for one variable**, never nest them, no side effects inside. Then the trap, worth running live:

```
cout << x > 0 ? "positive" : "negative";
```

```
error: no match for 'operator>' ('std::basic_ostream<char>' and 'int')
```

`<<` binds tighter, so the compiler read `(cout << x) > 0`. Parenthesise it inside any larger expression.

Re-run Monday's test. It still passes: after two appends the capacity is still 2, because doubling goes 0, 1, 2. **Our test cannot tell the two versions apart.** Worth asking the room what a passing test does and does not prove.

Beat 3. It leaks right now (10 min)

Talk to your neighbor. A `PyList` goes out of scope at the end of a function. What happens to the array it asked for?

Settle it by running it, not by saying it.

```
g++ -std=c++17 -g -fsanitize=address -o leak leak.cpp PyList.cpp  
&& ./leak
```

```
SUMMARY: AddressSanitizer: 20 byte(s) leaked in 1 allocation(s).
```

What one line fixes this, and where does it go?

Take both halves from the room; they wrote the line itself an hour ago.

```
PyList::~PyList() { delete [] myArray; myArray = nullptr; }
```

The compiler writes a destructor for you and it destroys the **members**. It destroys the pointer. It does not touch what the pointer points at.

If your class calls `new`, your class defines a destructor.

Re-run the sanitizer. Clean.

Beat 4. The debugger, on our own class (16 min)

Drive `PyList` on the projector, with them following on their own machines. Leave a05's `experiment` alone; they should meet that one cold.

1. **Breakpoint.** Gutter left of the line number, red dot, on `int cap = ...`.
2. **F5**, pick the configuration. It stops on the dot.
3. **Variables.** Expand `this`; add `myArray` to **Watch**.
4. **F10** over, **F11** into, **F5** continue.

Append nine items. Before each F5: will it stop again? Will the address change?

Run Monday's version first and let the prediction fail: it stops on every append with a new address. Today's stops five times, capacity 1, 2, 4, 8, 16.

Why do the stops get further apart?

The array moves, and anything holding the old address is holding a lie.
Close: Friday, these same four moves on a program that misbehaves.

Beat 5. What this unlocks (6 min)

You have written the thing that makes a list a list. Everything that grows works this way: ask for more room, copy across, release the old, repoint. Python's `list` does it, and so does C++'s `vector`.

Friday: hand a `PyList` to a function and see what the copy takes with it.